Republic of the Philippines
DEPARTMENT OF SCIENCE AND TECHNOLOGY
Philippine Atmospheric Geophysical Astronomical Services Administration

TenDay : Weather Forecast

API Documentation

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Overview

Welcome to the TenDay Weather Forecast API, a streamlined service built to provide 10-day weather forecasts across various locations in the Philippines. Designed for both developers and institutions, this API offers seamless access to forecast data at the regional, provincial, and municipal levels.

Key Features

- 10-Day Forecast Coverage: Includes daily information such as rainfall, wind, temperature, and more.
- Location-Based Queries: Easily retrieve forecast data by region, province, or municipality using names or PSGC codes.
- Smart Search: Supports flexible name matching and fuzzy search for easier location lookups.
- Detailed Metadata: Each API response contains version, timestamp, status code, and method for better context and traceability.

Common Use Cases

- Climate monitoring systems
- Disaster risk planning and early warning tools
- Agricultural and environmental platforms
- Public-facing weather forecast applications
- Data pipelines for visualizations and reports

Glossary of Terms

This section defines key terms and abbreviations used throughout the TenDay Weather Forecast API documentation

API *Application Programming Interface*	A set of rules and protocols that allow different software applications to communicate with each other. In this system, APIs let other tools or systems access climate data securely.
API Token	A unique string of characters used to authenticate and authorize a user or system to access the API. Tokens ensure secure communication.
Burst Limit	The maximum number of consecutive requests allowed before triggering a cooldown.
CERAM *Climate Exposure Risk and Adaptation Matrix*	A dataset or model used to analyze long-term climate projections and impacts.
Cooldown Period	A time window (e.g., 60 seconds) during which no requests are allowed after hitting the burst limit.
Forecast	A prediction or estimate of future weather or climate conditions based on models and historical data.
HTTP Status Code	A code returned by the API indicating success *(200),* client error *(400),* or server error *(500).*
PSGC *Philippine Standard Geographic Code*	Standardized numerical codes assigned to regions, provinces, and municipalities.
Rate Limiting	A method of controlling the number of requests a user can make within a certain time frame.

Getting Started

To begin using the TenDay Weather Forecast API, you'll need to complete a short authorization process to ensure secure and proper usage. This includes submitting a formal request, receiving approval, activating your token, and authenticating your API calls. Follow the steps below to request access, activate your credentials, and start integrating weather data into your applications.

Step 1: Prepare Your Request Letter
Draft an official request letter addressed to:
NATHANIEL T. SERVANDO
Administrator

Thru:
THELMA A. CINCO – *Project Leader, CIS4A&H*
MAXIMO F. PERALTA – *Chief, Engineering and Technical Services Division*

Your letter must include:

- The purpose of API use
- When and where the API will be used
- Your full name, email address, and phone number

Don't forget to fill out the required request form and review the guidelines.

Step 2: Wait for Approval
You will receive a notification via email regarding the status of your request.

If approved, the email will contain:

- Your API token
- A validation link

If declined, you may revise your submission and reapply.

Step 3: Activate Your API Access
Click the validation link provided in the approval email to activate your access.

Once activated:

- Your API token becomes valid
- You may begin making authenticated requests

Step 4: Authenticate Your Requests
Include your API token in the Authorization header of each request.

Example:
```
GET /api/tenday/date?province=Sorsogon&date=2025-07-08

Headers: {
  "token": "YOUR_API_TOKEN"
}
```

Step 5: Start Using the API

You're now ready to integrate TenDay forecast data into your application. Refer to the full documentation to explore available endpoints, query parameters, response formats, and data upload options.

Authentication

All requests to the API must include a valid token:

```
  "token": "YOUR_API_TOKEN"
```

Tokens are tied to specific users and must be kept secure. Tokens can be revoked at any time due to misuse or violation of terms.

Rate Limit

To ensure fair and stable access for all users, the TenDay Weather Forecast API enforces the following usage limits per registered API token:

Limit Type	Rule
Burst Limit	Maximum of 100 consecutive requests per token
Cooldown Period	A 60-second cooling time is enforced after hitting the burst limit
Daily Quota	Maximum of 1,000 requests per day per API

If you exceed 100 consecutive requests, your token will be temporarily restricted for 60 seconds before accepting new requests.

The daily quota resets at 00:00 PST.
Requests beyond the daily limit will result in a 429 Too Many Requests error.

Note: *If your application requires higher limits, you may request an upgrade by contacting the API support team with a justification.*

Error Codes

The API uses standard HTTP status codes to indicate request success or failure.

Status Code	Message	Description
200	OK	Request was successful
400	Bad Request	Invalid parameters or request format
401	Unauthorized	Missing or invalid API token
403	Forbidden	Access denied to the requested resource
404	Not Found	Requested resource or location not found
429	Too Many Requests	Rate limit exceeded
500	Internal Server Error	An unexpected server error occurred

API Structure

All successful responses from the TenDay Weather Forecast API follow a consistent JSON structure, typically composed of the following main parts:

data (Forecast Results)

This section contains the actual forecast information that was requested — such as rainfall, wind speed, temperature, etc., organized by location and date.

```
  "data": {
    "date": "7/7/25",
    "province": "Albay",
    "municity": "Tabaco City",
    "rainfall_desc": "LIGHT RAINS",
    "rainfall_total": 18.2,
```

```
    "cloud_cover": "CLOUDY",
    "tmean": 28.49,
    "tmin": 25.65,
    "tmax": 31.33,
    "humidity": 86,
    "wind_speed": 3.26,
    "wind_direction": "SSW",
  }
```

What it includes:

- Location-based forecast (province and municipality)
- Weather variables: rainfall, wind speed, temperature
- Target forecast date (date)

This section is the primary content users are requesting.

metadata (Forecast Dataset Information)

The metadata field provides descriptive information about the forecast dataset itself, such as what kind of forecast it is, which location it covers, and its issuance reference.

This section is shared publicly and helps users verify which dataset they are viewing.

```
  "metadata": {
    "request_no": 12345,
    "api": "Forecast by Date",
    "forecast": "10-day Forecast",
    "issuance_date": "7/7/2025",
    "region": " Bicol Region (Region V)",
    "province": "Camarines Norte",
    "municity": "Daet"
  }
```

Field	Description
`request_no`	A unique reference ID for the request or dataset
`Api`	The name of the API module used (Current Forecast, etc.)
`Forecast`	The type of forecast (e.g., 10-day Forecast)
`issuance_date`	The date when the forecast was officially released
`Region`	The full name of the region (with label)
`Province`	The province covered by the forecast
`Municity`	The municity covered by the forecast

This section helps users understand the scope and source of the forecast data they're viewing.

misc (Request Processing Info)

The misc object provides technical details about how the API handled the request. It includes request method information, response status, versioning, and pagination metadata (when applicable).

This object is primarily intended for developers and integrators to verify and debug API behavior.


```
"misc": {
  "version": "1.0",
  "timestamp": "7/7/2025 10:36:47 AM",
  "method": "GET",
  "current_page": 1,
  "per_page": 10,
  "total_count": 12,
  "total_page": 2,
  "status_code": 200,
  "description": "OK",
}
```


Field	Description
`Version`	API version used to process the request
`Timestamp`	Date and time when the request was handled (server time)
`Method`	HTTP method used (e.g., GET, POST)
`current_page`	Current page number in paginated results
`per_page`	Maximum number of records returned per page
`total_count`	Total number of matching records (string format for consistency)
`total_page`	Total number of pages based on the current query and pagination settings
`status_code`	HTTP response status code (200, 400, 404, etc.)
`Description`	Short message describing the result of the request

Ten-day Forecast API

Provides detailed short-term weather forecasts per municipality, including key parameters such as rainfall, temperature, humidity, and wind conditions over the next 10 days.

Current Forecast

Provides current weather parameters for each municipality or province.

```
https://tenday.pagasa.dost.gov.ph/api/v1/tenday/current
```

Parameter	Requirement	Description
municity	either required	Filters the forecast by municipality or city. ** For available values, refer to the Location API under Validation.*
province	either required	Filters the forecast by province. ** For the full list, check the Location API under Validation.*
region	either required	Filters the forecast by region. ** To view the complete list, refer to the Location API section under Validation.*
page	optional	Allows access to additional pages of the API response. Use none to disable pagination.

Full Forecast

Returns full 10-day weather data for each municipality or province.

```
https://tenday.pagasa.dost.gov.ph/api/v1/tenday/full
```

Parameter	Requirement	Description
municity	either required	Filters the forecast by municipality or city. ** For available values, refer to the Location API under Validation.*
province	either required	Filters the forecast by province. ** For the full list, check the Location API under Validation.*
region	either required	Filters the forecast by region. ** To view the complete list, refer to the Location API section under Validation.*
page	optional	Allows access to additional pages of the API response. Use **none** to disable pagination.

Date Forecast

Fetches weather data by municipality or province for a specific date

```
https://tenday.pagasa.dost.gov.ph/api/v1/tenday/date
```

Parameter	Requirement	Description
date	required	Filters forecast results based on the selected date. (format: MM-DD-YYYY) ** Refer to the Issuance API under 10-Day for available dates.*
municity	either required	Filters the forecast by municipality or city. ** For available values, refer to the Location API under Validation.*
province	either required	Filters the forecast by province. ** For the full list, check the Location API under Validation.*
region	either required	Filters the forecast by region. ** To view the complete list, refer to the Location API section under Validation.*
page	optional	Allows access to additional pages of the API response. Use **none** to disable pagination.

Issuance

Retrieves the most recent forecast issuance date and time.

```
https://tenday.pagasa.dost.gov.ph/api/v1/tenday/issuance
```

Seasonal Forecast API

Delivers long-term climate outlooks by province, highlighting projected trends in rainfall and temperature for upcoming months to aid in planning and decision-making.

Provincial Forecast

Provides a seasonal forecast for the next six months for each province.

```
https://tenday.pagasa.dost.gov.ph/api/v1/seasonal/province
```

Parameter	Requirement	Description
value	required	Specifies the type of data to retrieve: • pn – *Percent Normal (climatological comparison)* • mm – *Forecasted Rainfall (in millimeters)* • all – *Retrieves both pn and mm*

province	either required	Name of the province for which to retrieve the forecast. ** For the full list, check the Location API under Validation.*
page	optional	Allows access to additional pages of the API response. Use **none** to disable pagination.

Regional Forecast

Provides a seasonal forecast for the next six months per province, grouped by region.

```
https://tenday.pagasa.dost.gov.ph/api/v1/seasonal/region
```

Parameter	Requirement	Description
value	required	Specifies the type of data to retrieve: • pn – *Percent Normal (climatological comparison)* • mm – *Forecasted Rainfall (in millimeters)* • all – *Retrieves both pn and mm*
region	either required	Use region code to filter by region. • Numerical: *1, 2, 3* • Roman numerals: *i, ii, iii* • Alphanumeric: *4a, 4b* • Alphabetical: *ncr, car* • PSGC: *0500000000* ** For available values, refer to the Location API under Validation.*
page	optional	Allows access to additional pages of the API response. Use **none** to disable pagination.

Issuance

Retrieves the most recent forecast issuance date and time.

```
https://tenday.pagasa.dost.gov.ph/api/v1/seasonal/issuance
```

Climate Projections API

Provides future climate data (temperature and rainfall) based on global scenarios like SSP126, SSP245, and SSP585. Useful for planning, research, and risk assessment.

CERAM

Climate Exposure, Risk, and Adaptation Mapping — offers province-level climate risk data to support adaptation and vulnerability analysis.

```
https://tenday.pagasa.dost.gov.ph/api/v1/projections/ceram
```

Parameter	Requirement	Description
Province	optional	Filter by province name
Indicator_code	optional	Climate indicator code (e.g., temperature or rainfall metric)
Range	optional	Projection range or quantile
observed_baseline	optional	Observed baseline value used for comparison
Scenario	optional	Emissions scenario (based on SSPs)
start_period	optional	Starting year of projection
end_period	optional	Ending year of projection
Page	optional	Allows access to additional pages of the API response. Use **none** to disable pagination.

Files API

Allows you to download different types of forecast files by date, including ten-day, seasonal, and other forecast data. Useful for accessing and organizing files for analysis or reporting.

10-day File Retrieval API

Lets you download weather forecast files for up to 10 days. You can get a single day's file or all 10 days at once, making it easy to access and use the data.

```
https://tenday.pagasa.dost.gov.ph/api/v1/file/tenday
```

Parameter	Requirement	Description
issuance_date	required	The date the forecast was issued. (format: YYYYMMDD) ** For the latest issuance date, check the Issuance API under 10-day.*

file	required	Specifies the forecast file to retrieve. • TMEAN – *Mean Temperature* • TMIN – *Minimum Temperature* • TMAX – *Maximum Temperature* • RH – *Relative Humidity* • TCC – *Total Cloud Cover* • TP – *Total Precipitation* • WD – *Wind Direction* • WS – *Wind Speed*
token	required	API token for authentication.
target	optional	Specifies the forecast file date you want to download (format: YYYYMMDD) ** To determine the target dates, use the start and end dates from the 10-day Issuance API as your reference range.*
masked	optional	If **"true" or "1"**, returns the masked version of the file (e.g., clipped to country boundaries). Default is **"false" or "0"**.

Seasonal File Retrieval API

Lets you download weather forecast files for up to 10 days. You can get a single day's file or all 10 days at once, making it easy to access and use the data.

```
https://tenday.pagasa.dost.gov.ph/api/v1/file/seasonal
```

Parameter	Requirement	Description
batch	required	The batch number representing the forecast period ** For the latest batch, check the Issuance API under Seasonal* .
type	required	Specifies which forecast file to retrieve: • PN – Percent Normal • MM – Forecast Rainfall • ALL – Both
token	required	API token for authentication.

CERAM File Retrieval API

Lets you download datasets used for the Climate Extremes Risk Analysis Matrix (CERAM). These files support analysis of extreme climate indicators under various future climate scenarios.

```
https://tenday.pagasa.dost.gov.ph/api/v1/file/ceram
```

Parameter	Requirement	Description
token	required	API token for authentication.
climate_indicator	required	Main climate variable to filter data. • RR – Rainfall • TMAX – Max Temperature • TMIN – Min Temperature
indicator_code	optional	Specific indicator under the selected climate variable: • RR: rx1day, rx5day • TMAX: txm, txn, txx • TMIN: tnm, tnn, tnx
percentile	optional	Statistical level of the dataset. • 10, 25, 50, 75, 90, mean
ssp	optional	Shared Socioeconomic Pathway scenario. • 119, 126, 245, 370, 585

Validation API

Used to verify input values such as PSGC codes, indicator codes, and climate scenarios to ensure parameters are correct before making data requests. Also allows validation of your API token to confirm if it's active and properly authorized.

Validate API

Checks if the provided API token is valid, shows which APIs the token is authorized to access, and returns the token’s expiration date. Useful for verifying access rights and session status before making data requests.

```
https://tenday.pagasa.dost.gov.ph/api/v1/validate
```

Parameter	Requirement	Description
token	required	API token for authentication.

Location API

Provides a list of regions, provinces, and municipalities with corresponding PSGC codes. Supports location-based filtering for forecast and projection data.

```
https://tenday.pagasa.dost.gov.ph/api/v1/location
```

Parameter	Requirement	Description
region	optional	Filter results by region
province	optional	Filter results by province